Chapter 2: Electrostatic Potential and Capacitance

Formula Sheet + Tips & Tricks

Formula Sheet

I. Electric Potential and Potential Difference

Potential due to a point charge

$$V = \frac{kQ}{r}$$

$V \propto 1/r$; slope of a V vs $1/r$ graph gives kQ

Potential difference & work

$$V = \frac{W}{q}; \quad W = q(V_B - V_A)$$

W = work done in moving charge q between two points

Potential due to several charges

$$V = \sum_i \frac{kQ_i}{r_i}$$

scalar sum (superposition), unlike vector sum for $\vec{E}$

Kinetic energy gained

$$K = qV$$

gain in KE = work done by the field for a charge accelerated through PD V

II. Potential Gradient

Potential gradient

$$E = \frac{\Delta V}{d}$$

electric field = (negative of) rate of change of potential with distance; potential decreases along $\vec{E}$

III. Electrostatic Potential Energy

PE of two point charges

$$U = \frac{kq_1q_2}{r}$$

energy needed to assemble the charge pair from infinity

Energy in electron-volt

$$1\ \text{eV} = 1.6 \times 10^{-19}\ \text{J}$$

convenient energy unit at atomic scale

IV. Electric Dipole (Energy Aspects)

Dipole moment

$$p = q \times 2l$$

$2l$ = separation between the two charges of the dipole

Torque on a dipole

$$\tau = pE\sin\theta$$

maximum at $\theta = 90°$: $\tau_{max} = pE$

PE of a dipole in a field

$$U = -pE\cos\theta$$

$U = 0$ at $\theta = 90°$; minimum $(-pE)$ at $\theta = 0°$

Work done rotating a dipole

$$W = pE(\cos\theta_1 - \cos\theta_2)$$

work to rotate dipole from θ_1 to θ_2 in the field

V. Capacitance of a Conductor

Capacitance

$$C = \frac{Q}{V}$$

ratio of charge to potential; unit: farad (F)

Capacitance of an isolated sphere

$$C = 4\pi\varepsilon_0 R = \frac{R}{k}$$

R = radius of the sphere

VI. Parallel Plate Capacitor

Parallel plate capacitance (air)

$$C = \frac{\varepsilon_0 A}{d}$$

A = plate area, d = plate separation

Charge on capacitor

$$Q = CV$$

V = potential difference applied across the plates

VII. Capacitor with Dielectric Slab

Capacitance with dielectric

$$C = \frac{K\varepsilon_0 A}{d}$$

K = dielectric constant of the medium between the plates

Effect of dielectric on C, Q, V

$$Q = \text{const if battery removed; } V = \text{const if battery connected}$$

battery removed $\Rightarrow V$ falls, Q fixed; battery connected $\Rightarrow V$ fixed, Q rises

VIII. Energy of a Charged Capacitor

Energy stored

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV$$

use whichever form matches the known quantities (Q fixed or V fixed)

Energy density

$$u = \frac{U}{\text{Volume}} = \frac{1}{2}\varepsilon_0 E^2$$

energy stored per unit volume of the field region

IX. Combination of Capacitors

Series combination

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots$$

same charge Q on each capacitor; PDs add up

Parallel combination

$$C_p = C_1 + C_2 + \cdots$$

same PD across each capacitor; charges add up

X. Miscellaneous

Potential of a charged sphere

$$V_{surface} = \frac{kQ}{R}; \quad V_{inside} = V_{surface} \text{ (constant)}$$

for a hollow/solid conducting sphere, potential inside equals that on the surface

Balancing a charged drop

$$qE = mg$$

condition for a charged drop to remain stationary between plates

Tips, Tricks & Hacks

- $V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$ is a scalar — superpose potentials by simple algebraic addition (with sign), unlike fields which need vector addition.
- Potential is constant on an equipotential surface, and equipotentials are always $\perp$ to field lines — use this to sketch quickly.
- $E = -\frac{dV}{dr}$: if V vs r graph is given, the field is (minus) the slope — a favourite graph-based MCQ.
- For capacitors in series, charge Q is the same on each; in parallel, voltage V is the same on each. Memorise this pairing to avoid mixing up formulas.
- Energy stored $U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}$ — pick the version that matches what's held constant (battery connected $\Rightarrow$ V fixed; battery disconnected $\Rightarrow$ Q fixed) before and after inserting a dielectric/changing plate separation.
- Dielectric slab fully filling the gap: $C' = KC_0$. Partially filling: treat as series combination of air gap and dielectric gap.
- Common trap: after charging and then *disconnecting* the battery, Q stays constant, not V; if the battery stays connected, V stays constant, not Q. Re-derive energy change from whichever is fixed.
- Work done in moving a charge on the *same* equipotential surface is always zero, regardless of the path.